





Submitted to-

Dr. Raihan Ul Islam

Associate Professor

Department of Computer Science and Engineering

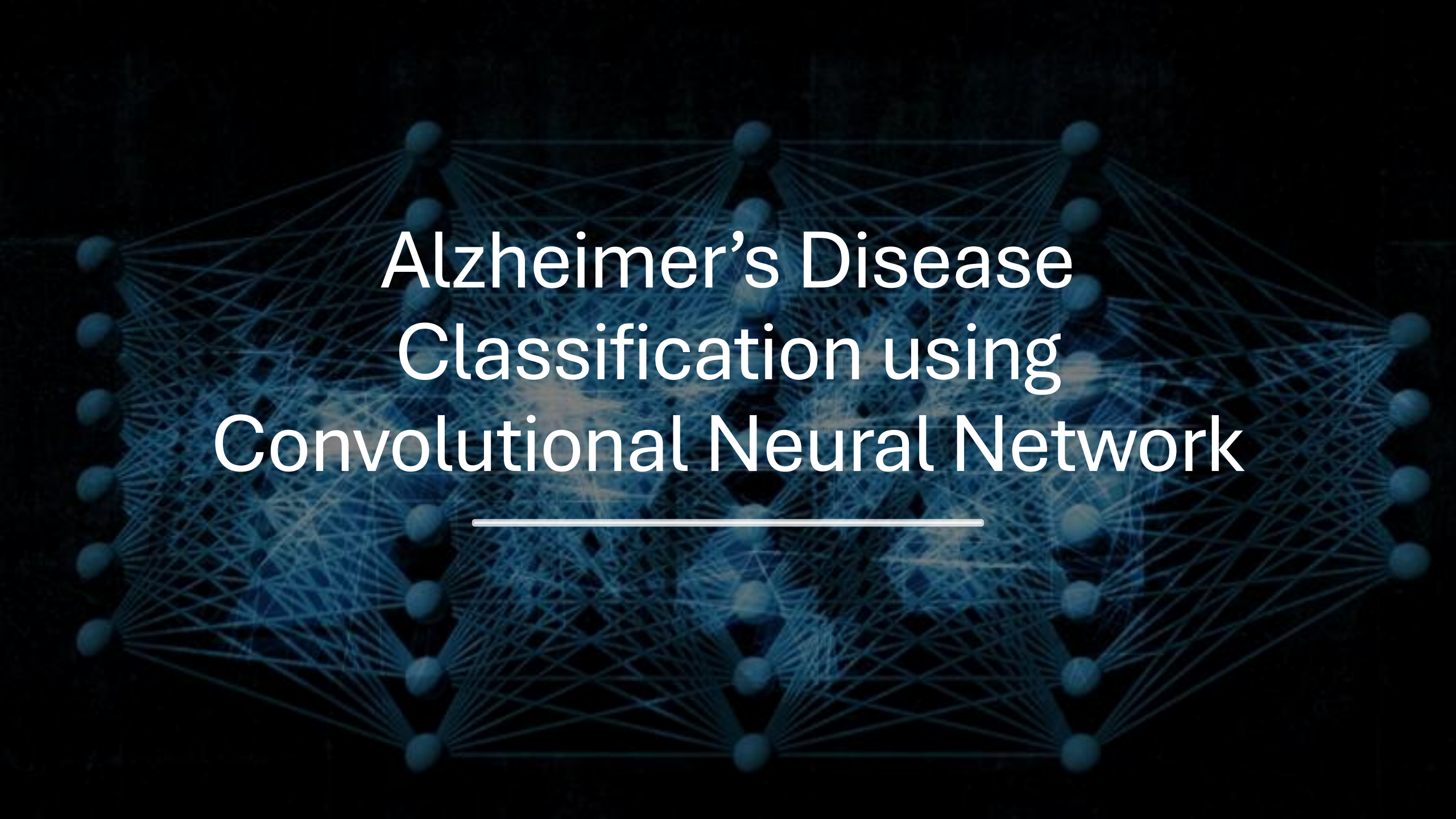
Submitted by-

Name: Mujahidul Islam

ID : 2020-2-60-190

Name: Nafiz Khan

ID: 2020-2-60-175



Alzheimer's Disease Classification using Convolutional Neural Network

All about Alzheimer's

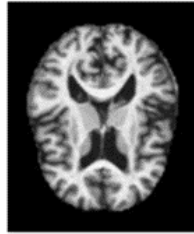
- * What is Alzheimer's disease
- * Different stages

About Dataset

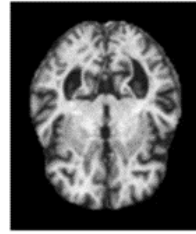
Alzheimer's dataset (MRI segmentation)

Classes :

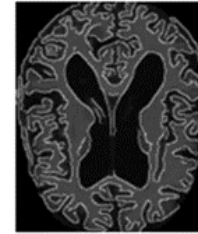
1. Mild Demented
2. Very Mild Demented
3. Moderate Demented
4. Non Demented



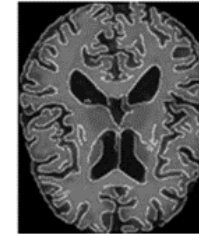
Very Mild Demented
(Raw Image)



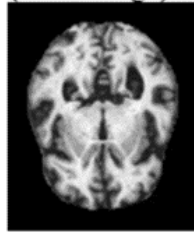
Very Mild Demented
(Raw Image)



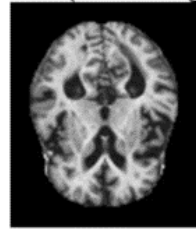
Very Mild Demented
(Processed Image)



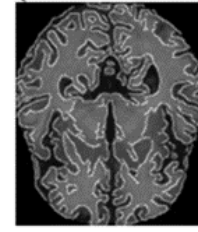
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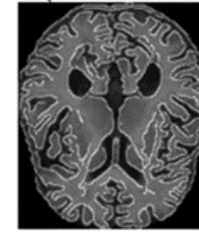
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(Raw Image)



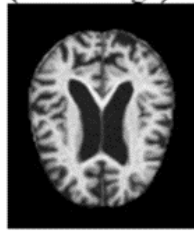
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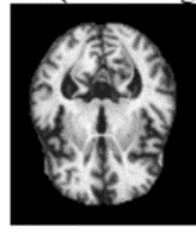
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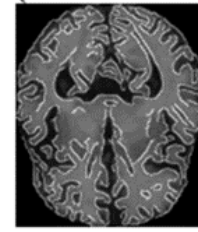
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(Processed Image)



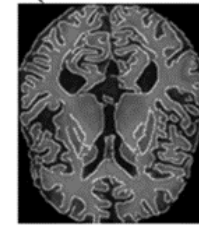
Moderate Demented
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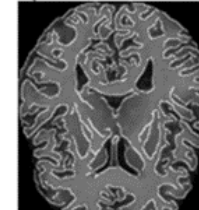
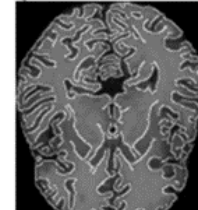
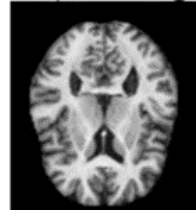
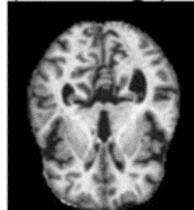
Moderate Demented
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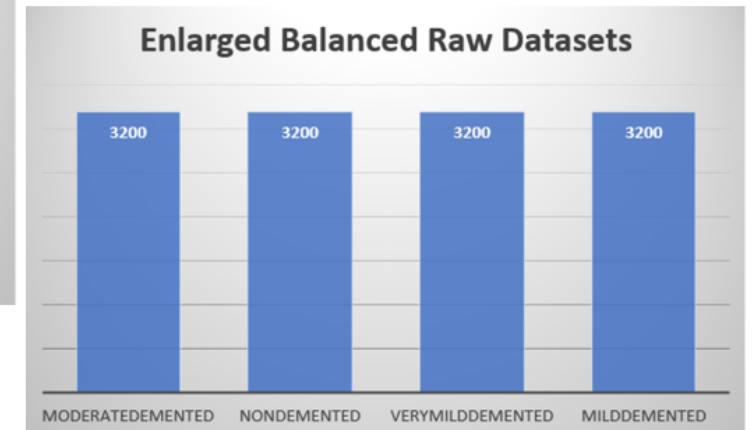
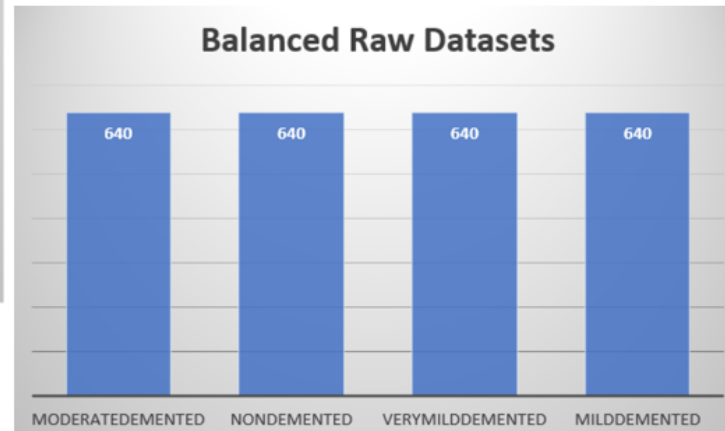
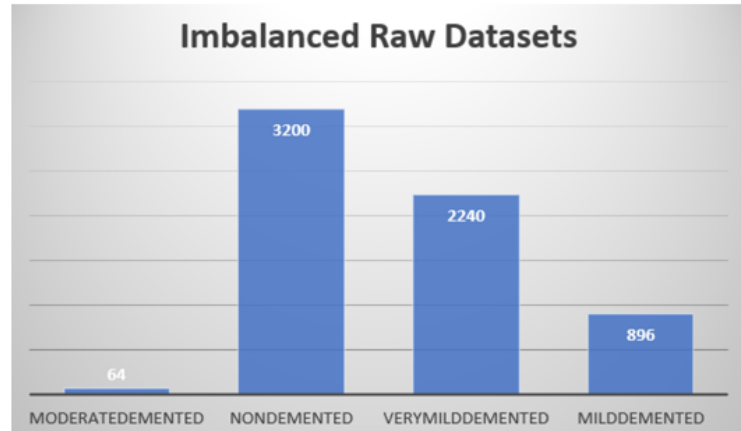
Moderate Demented
(Processed Image)



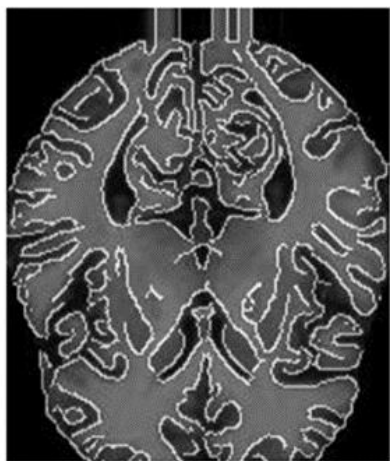
Moderate Demented
(Processed Image)



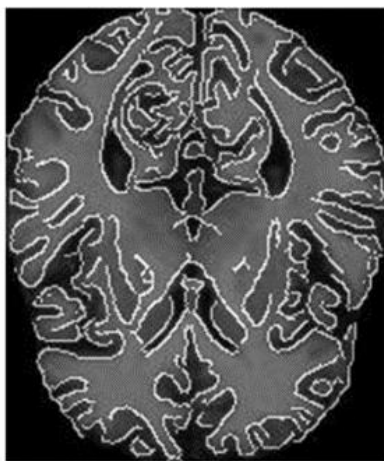
Data Collection



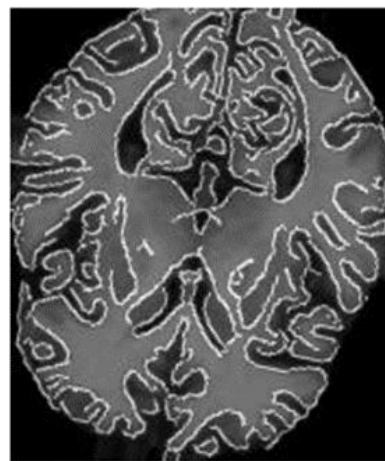
Ready for Pre-Processing



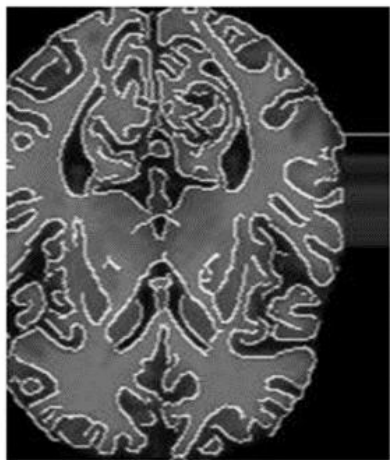
Height Shift



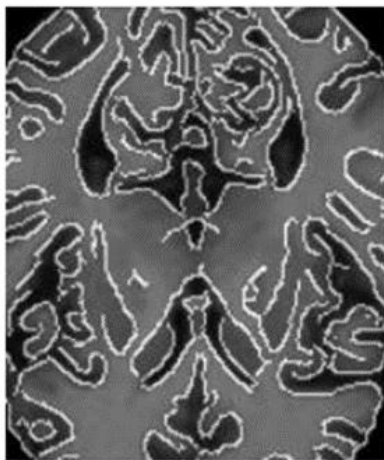
Horizontal Flip



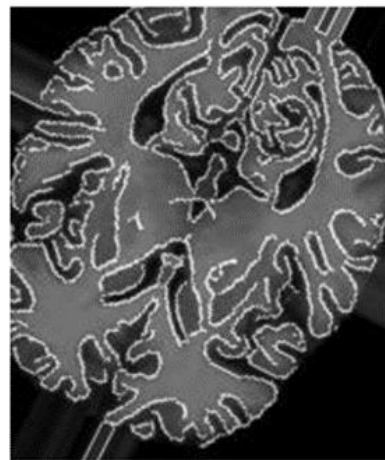
Rotated 20*



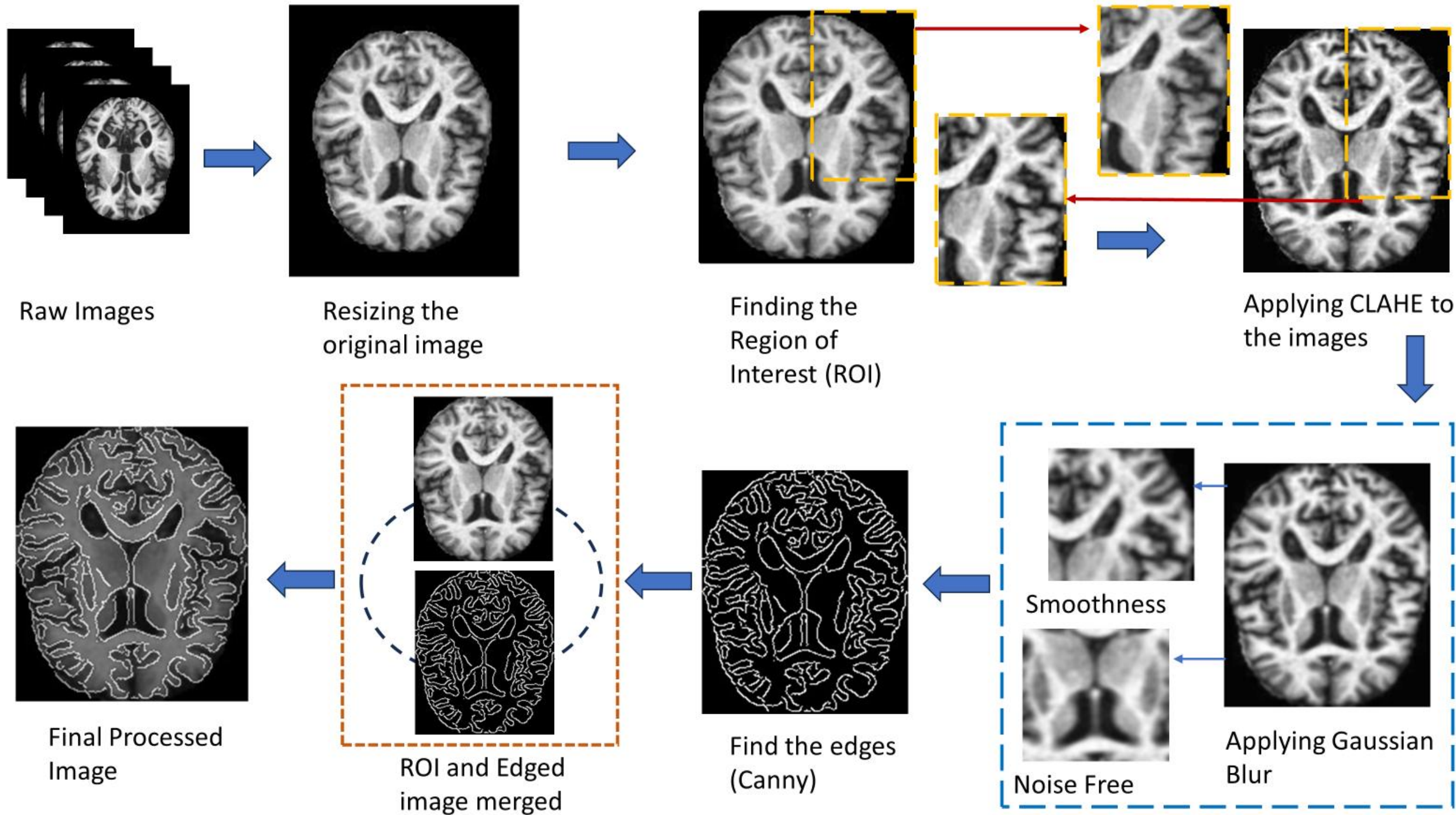
Width shift



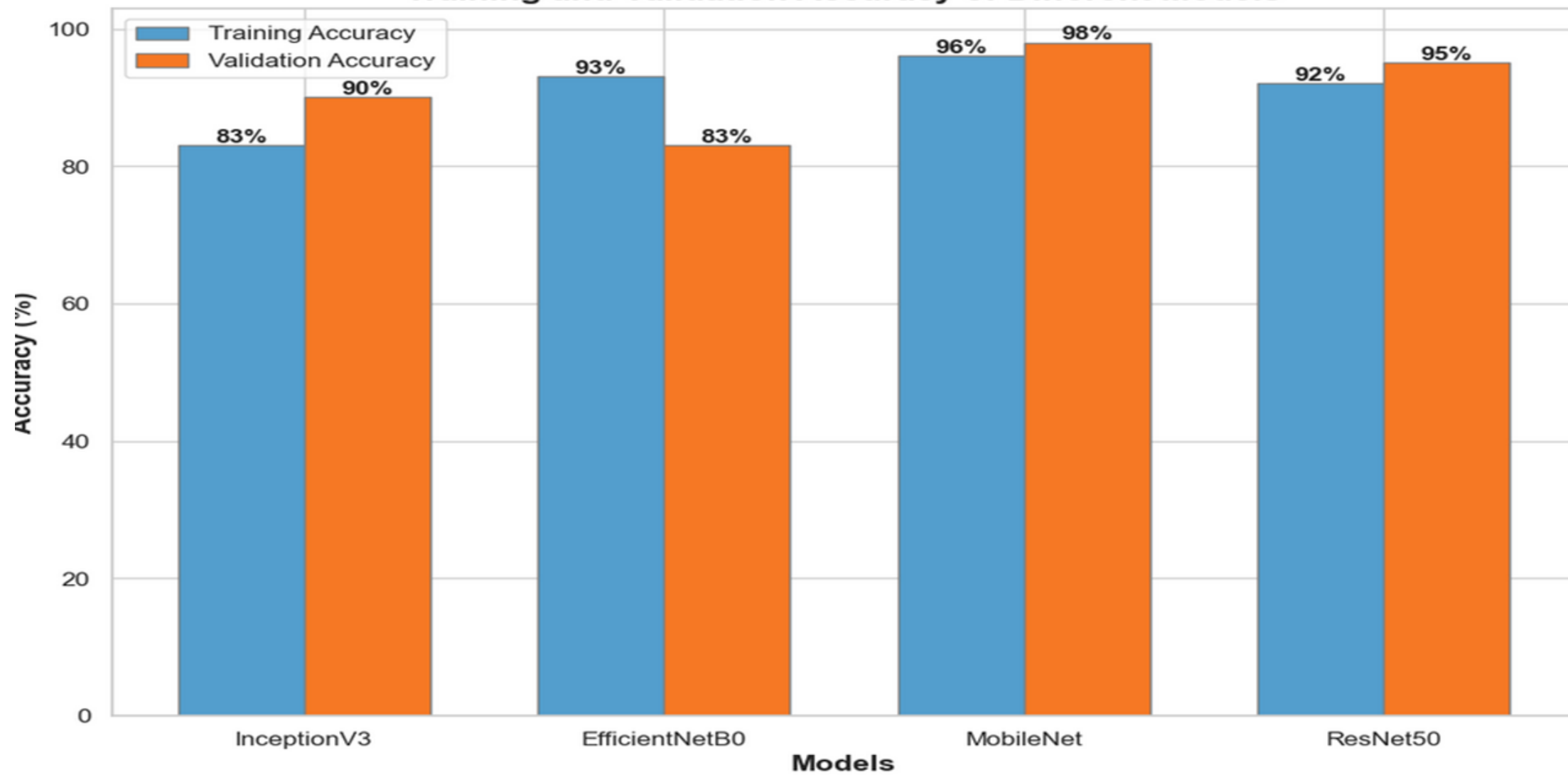
Zoom

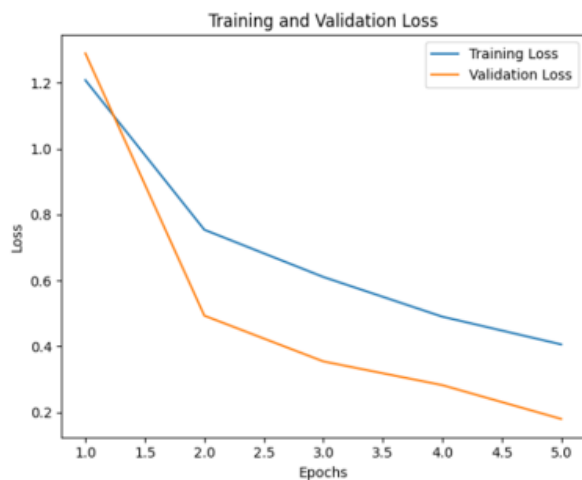


Rotated 30*

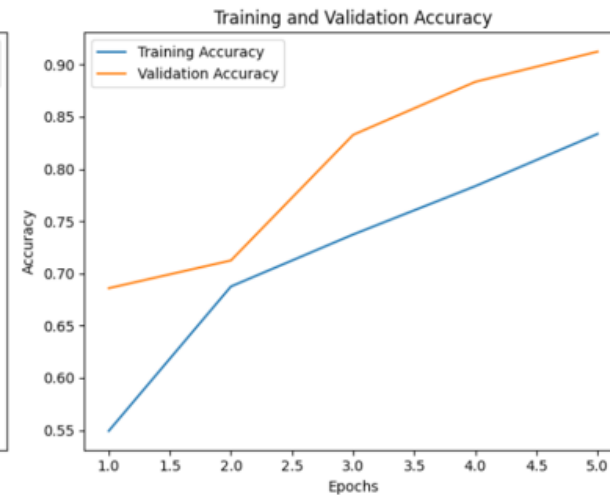
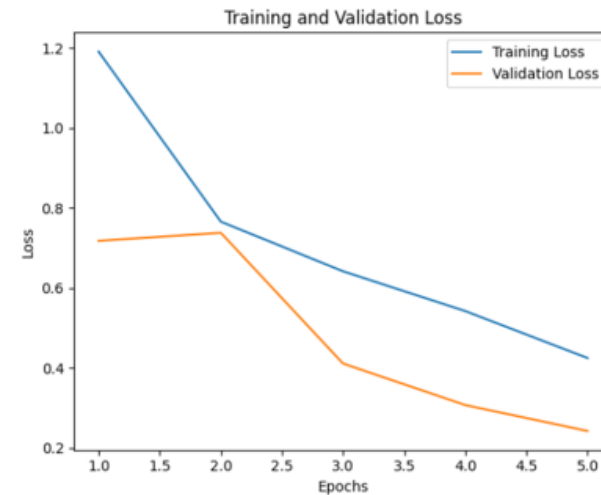


Training and Validation Accuracy of Different Models

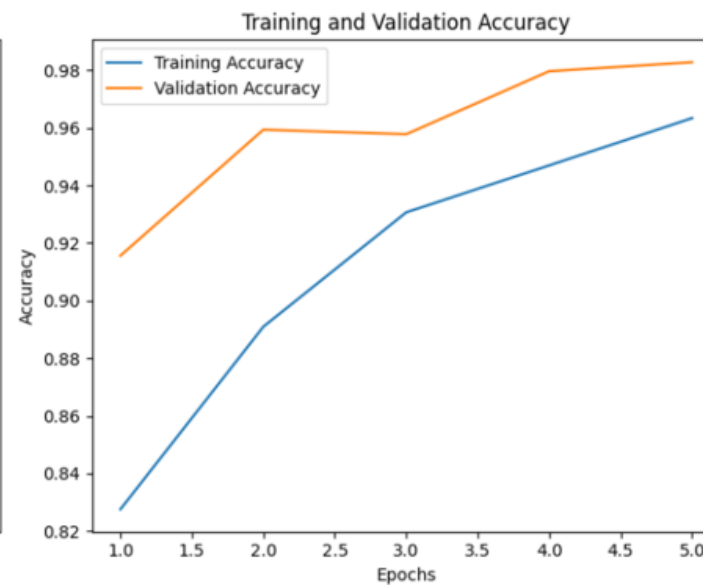
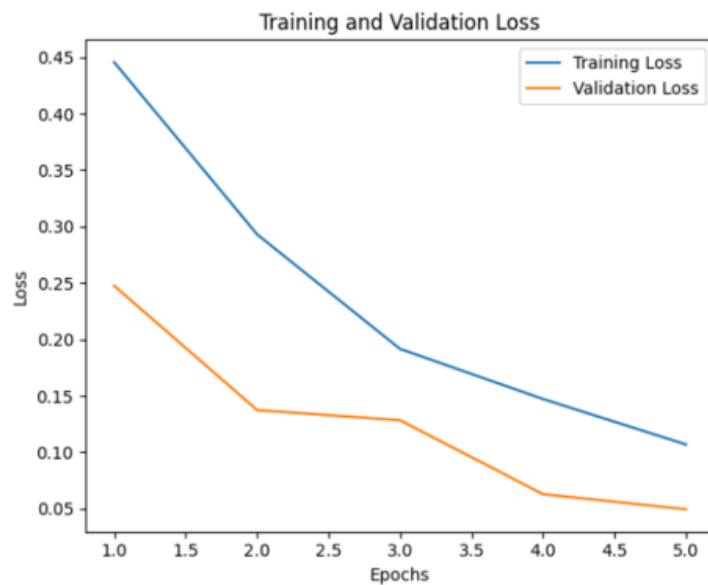




EfficientNetB0 Model



InceptionV3 Model



MobileNet Model



**Any
Questions?**



